SCHNEIDER ELECTRIC

INDUSTRIAL QUALITY & RELIABILITY ENGINEERING

8th October | Quality Engineer — Jagriti Singh | Day 73 (100 Days Quality Management Challenge)

DISCLAIMER: This is an independent educational case study created as part of my "100 Days Quality Management Challenge." Schneider Electric is a separate organisation. This is not an official Schneider Electric project, report, collaboration, or representation. The analysis applies general Quality Engineering principles using publicly available information and does not use confidential company information.

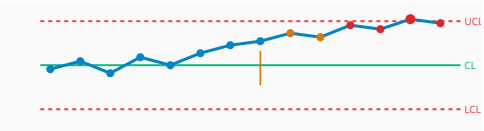
QUALITY TARGET

Critical electrical and industrial products, with focus on manufacturing process consistency, product reliability, supplier quality and prevention-based quality.

QUALITY PROBLEM

- Process data may remain within specification while showing gradual mean shift or increasing variation.
- Critical supplier components may meet final inspection requirements but show intermittent field failures.
- Borderline process capability can increase future defect and reliability risk.
- Corrective actions must be verified for sustained effectiveness, not only short-term improvement.
- Design changes require assessment of both manufacturing capability and reliability risk.

PROCESS & RELIABILITY DIAGNOSTICS



EXPECTED IMPACT

- Improved process stability
- Reduced process variation
- Earlier detection of risks
- Improved supplier reliability
- Reduced recurring failures
- Better CAPA effectiveness
- Prevention-based quality
- Predictable manufacturing

QUALITY ENGINEERING TOOLS

SPC & Trend Analysis
Monitor process stability, mean shifts and variation.

Capability — Cp/Cpk
Evaluate process spread and centering vs spec limits.

FMEA
Identify and prioritize potential process/ product risks.

FTA / Failure Analysis
Investigate recurring or intermittent failure mechanisms.

RCA
Identify underlying causes rather than treating symptoms.

CAPA / Corrective Action
Implement and control robust corrective actions.

Reliability Testing & Verification
Confirm that improvements and design changes perform reliably in operational environments.

PROPOSED QUALITY APPROACH

Monitor → Detect → Analyse → Risk → Correct → Verify → Sustain

1. **Monitor:** Use SPC & trend data to detect process shifts/variation.
2. **Detect:** Identify abnormal trends, failures or capability concerns.
3. **Analyse:** Evaluate process capability, failure patterns and risk.
4. **Identify Risk:** Apply FMEA/FTA/RCA to determine root causes.
5. **Correct:** Implement targeted corrective & preventive actions.
6. **Verify:** Re-test process capability, reliability and effectiveness.
7. **Sustain:** Continue monitoring to ensure long-term stability.

"Quality is not only about meeting specifications. It is about controlling variation, managing risk and verifying that improvements remain effective over time."

My approach to the case:

Data → Variation → Risk → Root Cause → Corrective Action → Verification → Prevention

This quality solution is developed based on my own knowledge and understanding of Quality Engineering principles. I have used the tools, concepts and approaches that I have learned and practiced to analyse the problem and propose this solution.

Industry: Energy Management • Industrial Automation • Electrification
Quality Focus: Process Quality • Reliability • Supplier Quality • Risk Management • Continuous Improvement
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